

Seth Ricks

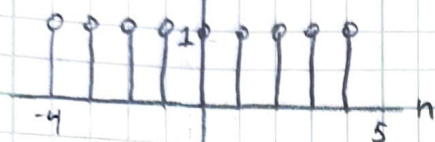
ECEN 380
3/9/2026

PS09: 3.61(b), 3.62(a),
3.68(a)(c), 3.70(a)(d),
MATLAB 3.117

3.61(b)

Use the table of transforms and properties to find the DTFT of the following signal:

$$x[n] = (n-2)(u[n+4] - u[n-5])$$



$$\Rightarrow \begin{cases} 1, & -4 \leq n \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

$$\boxed{1/6}$$

$$x[n] = (n-2) \cdot \begin{cases} 1, & |n| \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

$$C.7: -jn x[n] \xleftrightarrow{\text{DTFT}} \frac{d}{d\omega} X(e^{j\omega})$$

$$(n) x[n] \xleftrightarrow{\text{DTFT}} \frac{d}{d\omega} X(e^{j\omega}) \cdot \frac{1}{-j} = j \frac{d}{d\omega} X(e^{j\omega})$$

$$C.3: x[n] = \begin{cases} 1, & |n| \leq M \\ 0, & \text{otherwise} \end{cases} \xleftrightarrow{\text{DTFT}} X(e^{j\omega}) = \frac{\sin[\omega(\frac{2M+1}{2})]}{\sin(\omega/2)}$$

$$C.7: x[n-n_0] \xleftrightarrow{\text{DTFT}} e^{-j\omega n_0} X(e^{j\omega})$$

$$C.7: ax[n] + by[n] \xleftrightarrow{\text{DTFT}} aX(e^{j\omega}) + bY(e^{j\omega})$$

$$x[n] = n \cdot \begin{cases} 1, & |n| \leq 4 \\ 0, & \text{otherwise} \end{cases} = 2 \cdot \begin{cases} 1, & |n| \leq 4 \\ 0, & \text{otherwise} \end{cases} \cdot \begin{cases} 1, & |n| \leq 4 \\ 0, & \text{otherwise} \end{cases} \approx w[n]$$

$$W(e^{j\omega}) = \frac{\sin(\frac{9}{2}\omega)}{\sin(\omega/2)}$$

$$\text{DTFT}\{nw[n]\} = j \frac{d}{d\omega} \frac{\sin(\frac{9}{2}\omega)}{\sin(\omega/2)}$$

$$\text{DTFT}\{-2w[n]\} = -2 \frac{\sin(\frac{9}{2}\omega)}{\sin(\omega/2)}$$

$$\boxed{\text{DTFT}\{x[n]\} = X(e^{j\omega}) = j \frac{d}{d\omega} \left[\frac{\sin(\frac{9}{2}\omega)}{\sin(\omega/2)} \right] - 2 \cdot \frac{\sin(\frac{9}{2}\omega)}{\sin(\omega/2)}}$$

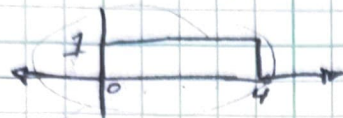
3.62(a)

Use the FT pair:

$$x(t) = \begin{cases} 1, & |t| < 1 \\ 0, & \text{otherwise} \end{cases} \xleftrightarrow{\text{FT}} X(j\omega) = \frac{2\sin(\omega)}{\omega}$$

$$\boxed{2/6}$$

and the FT properties to evaluate the frequency-domain representation of the signal depicted:



$$y(t) = x(t/4), \text{ where } x(t) = \begin{cases} 1, & |t| < 1 \\ 0, & \text{otherwise} \end{cases}$$

$$\text{C.7: } x(at) \xleftrightarrow{\text{FT}} \frac{1}{|a|} X\left(\frac{j\omega}{a}\right)$$

$$X(j\omega) = \frac{2\sin(\omega)}{\omega} \implies \text{FT}\{x(t/4)\} = \frac{1}{|1/4|} \cdot \frac{2\sin(\omega/4)}{\omega/4}$$

$$Y(j\omega) = 4 \cdot \frac{2\sin(4\omega)}{4\omega} = \frac{2\sin(4\omega)}{\omega}$$

$$\boxed{Y(j\omega) = \frac{2\sin(4\omega)}{\omega}}$$

3.68(a)

Determine the frequency and the impulse response for the system described by the following differential equation:

$$(a) \quad \frac{d}{dt} y(t) + 3y(t) = x(t)$$

3/6

class slides:

$$FT \{ h(t) \} = H(j\omega)$$

↑
impulse
response

↑
freq.
response

;

$$y(t) = h(t) * x(t)$$

$$Y(j\omega) = H(j\omega) X(j\omega)$$

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)}$$

$$C.7: \quad \frac{d}{dt} x(t) \xrightarrow{FT} j\omega X(j\omega)$$

$$(j\omega) Y(j\omega) + 3Y(j\omega) = X(j\omega)$$

$$(j\omega + 3) Y(j\omega) = X(j\omega)$$

$$\frac{Y(j\omega)}{X(j\omega)} = \frac{1}{j\omega + 3} = H(j\omega)$$

$$C.4: \quad x(t) = e^{-at} u(t), \operatorname{Re}\{a\} > 0 \xrightarrow{FT} X(j\omega) = \frac{1}{a + j\omega}$$

$$\operatorname{Inv. FT} \{ H(j\omega) \} \stackrel{\text{inv. FT}}{=} \frac{1}{3 + j\omega} = e^{-3t} u(t) = h(t)$$

Impulse response: $h(t) = e^{-3t} u(t)$

Frequency response: $H(j\omega) = \frac{1}{j\omega + 3}$

3.68(c)

Determine frequency & impulse responses for the system described by the following difference equation:

$$(c) \quad y[n] - \frac{1}{4}y[n-1] - \frac{1}{8}y[n-2] = 3x[n] - \frac{3}{4}x[n-1]$$

$$\text{DTFT}\{h[n]\} = H(e^{j\omega}) ; \quad y[n] = h[n] * x[n]$$

$$Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega})$$

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})}$$

$$\text{C.7: } x[n-n_0] \xrightarrow{\text{DTFT}} e^{-j\omega n_0} X(e^{j\omega})$$

$$Y(e^{j\omega}) - \frac{1}{4}e^{-j\omega}Y(e^{j\omega}) - \frac{1}{8}e^{-j2\omega}Y(e^{j\omega}) = 3X(e^{j\omega}) - \frac{3}{4}e^{-j\omega}X(e^{j\omega})$$

$$Y(e^{j\omega}) \left[1 - \frac{1}{4}e^{-j\omega} - \frac{1}{8}e^{-j2\omega} \right] = X(e^{j\omega}) \left[3 - \frac{3}{4}e^{-j\omega} \right]$$

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{3 - \frac{3}{4}e^{-j\omega}}{1 - \frac{1}{4}e^{-j\omega} - \frac{1}{8}e^{-j2\omega}} = \frac{3 - \frac{3}{4}e^{-j\omega}}{-\frac{1}{8}(e^{-j\omega} + 4)(e^{-j\omega} - 2)}$$

$$\begin{aligned} & \frac{-\frac{1}{8}x^2 - \frac{1}{4}x + 1}{-\frac{1}{8}(x^2 + 2x - 8)} = \frac{-24 + 6e^{-j\omega}}{(e^{-j\omega} + 4)(e^{-j\omega} - 2)} = \frac{A}{(e^{-j\omega} + 4)} + \frac{B}{(e^{-j\omega} - 2)} \\ & \frac{-\frac{1}{8}(x+4)(x-2)}{-\frac{1}{8}(x+4)(x-2)} \quad (x = e^{-j\omega} \text{ for now}) \end{aligned}$$

$$\frac{-24 + 6x}{(x+4)(x-2)} = \frac{A}{(x+4)} + \frac{B}{(x-2)} \Rightarrow -24 + 6x = A(x-2) + B(x+4)$$

$$6x + -24 = Ax - 2A + Bx + 4B$$

$$A + B = 6$$

$$-2A + 4B = -24$$

$$\begin{bmatrix} 1 & 1 & 6 \\ -2 & 4 & -24 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & -2 \end{bmatrix}$$

$$= \frac{8}{(e^{-j\omega} + 4)} + \frac{-2}{(e^{-j\omega} - 2)}$$

$$\text{C.3: } x(e^{j\omega}) = \frac{1}{1 - \alpha e^{j\omega}} \xrightarrow{\text{DTFT}} x[n] \alpha^n u[n], |\alpha| < 1$$

$$H(e^{j\omega}) = \frac{8}{(e^{j\omega} - 4) + (e^{j\omega} - 2)} + \frac{8}{4(1 + \frac{1}{4}e^{j\omega})} = \frac{8}{4(1 - (-\frac{1}{4}e^{j\omega}))} \quad \boxed{5/6}$$

$$= \frac{2}{1 - (-\frac{1}{4}e^{j\omega})}$$

$$\frac{-2}{2(1 - \frac{1}{2}e^{j\omega})} = \frac{1}{1 - \frac{1}{2}e^{j\omega}}$$

$$H(e^{j\omega}) = \frac{2}{1 - (-\frac{1}{4}e^{j\omega})} + \frac{1}{1 - \frac{1}{2}e^{j\omega}}$$

$$h[n] = \text{Inv. DTFT} \{ \sqrt{\quad} \} = 2\left(\frac{1}{4}\right)^n u[n] + \left(\frac{1}{2}\right)^n u[n]$$

$$H(e^{j\omega}) = \frac{2}{1 - (-\frac{1}{4}e^{j\omega})} + \frac{1}{1 - \frac{1}{2}e^{j\omega}} \quad (\text{frequency response})$$

$$h[n] = 2\left(\frac{1}{4}\right)^n u[n] + \left(\frac{1}{2}\right)^n u[n] \quad (\text{impulse response})$$

3.70(a) Determine differential-equation description for system w/ frequency response:

6/6

$$H(j\omega) = \frac{2 + 3j\omega - 3(j\omega)^2}{1 + 2j\omega}$$

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{2 + 3j\omega - 3(j\omega)^2}{1 + 2j\omega}$$

$$Y(j\omega)(1 + 2j\omega) = X(j\omega)(2 + 3j\omega - 3(j\omega)^2)$$

$$Y(j\omega) + 2j\omega Y(j\omega) = 2X(j\omega) + 3j\omega X(j\omega) - 3(j\omega)^2 X(j\omega)$$

$$\boxed{y(t) + 2 \frac{d}{dt} y(t) = 2x(t) + 3 \frac{d}{dt} x(t) - 3 \frac{d^2}{dt^2} x(t)}$$

3.70(d) Determine difference-equation description for system w/ frequency response:

$$H(e^{j\omega}) = \frac{1 + e^{-j\omega}}{e^{-j2\omega} + 3}$$

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{1 + e^{-j\omega}}{e^{-j2\omega} + 3}$$

$$Y(e^{j\omega})(e^{-j2\omega} + 3) = X(e^{j\omega})(1 + e^{-j\omega})$$

$$e^{-j2\omega} Y(e^{j\omega}) + 3Y(e^{j\omega}) = X(e^{j\omega}) + e^{-j\omega} X(e^{j\omega})$$

$$\boxed{y[n-2] + 3y[n] = x[n] + x[n-1]}$$

(3.117)

The frequency response of either a continuous or discrete time system described in state-variable form (see Problem 3.87) may be computed with the use of the MATLAB command `freqresp`. The syntax is **`h = freqresp(sys,w)`**, where **`sys`** is the object containing the state-variable description (see Section 2.14) and **`w`** is a vector containing the frequencies at which to evaluate the frequency response. Note that **`freqresp`** applies in general to multiple-input, multiple-output systems, so the output **`h`** is a multidimensional array. For the class of single-input, single-output systems considered in this text and **`N`** frequency points in **`w`**, **`h`** is of dimension 1 by 1 by **`N`**. The command **`squeeze(h)`** converts **`h`** to an **`N`**-dimensional vector that may be displayed with the **`plot`** command. Thus, we may obtain the frequency response by using the following commands:

```
>> h = freqresp(sys,w);  
>> hmag = abs(squeeze(h));  
>> plot(w,hmag)  
>> title('System Magnitude Response');  
>> xlabel('Frequency (rads/sec)'); ylabel('Magnitude');
```

Use MATLAB to plot the magnitude and phase response for the systems with state-variable descriptions given in Problems 3.88 and 3.89.

(Problem 3.88)

(a) $A = \begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix}; b = \begin{bmatrix} 0 \\ 2 \end{bmatrix}; c = [1 \quad 1]; D = [0]$

(b) $A = \begin{bmatrix} 1 & 2 \\ -3 & -4 \end{bmatrix}; b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}; c = [0 \quad 1]; D = [0]$

(Problem 3.89)

(a) $A = \begin{bmatrix} -1/2 & 1 \\ 0 & 1/4 \end{bmatrix}; b = \begin{bmatrix} 0 \\ 1 \end{bmatrix}; c = [1 \quad 0]; D = [1]$

(b) $A = \begin{bmatrix} 1/4 & 3/4 \\ 1/4 & -1/4 \end{bmatrix}; b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}; c = [0 \quad 1]; D = [0]$

```

A = [-2, 0; 0, -1];
b = [0; 2];
c = [1, 1];
D = 0;
sys1 = ss(A,b,c,D,-1);

A = [1, 2; -3, -4];
b = [1; 2];
c = [0, 1];
D = 0;
sys2 = ss(A,b,c,D,-1);

A = [-1/2, 1; 0, 1/4];
b = [0; 1];
c = [1, 0];
D = 1;
sys3 = ss(A,b,c,D,-1);

A = [1/4, 3/4; 1/4, -1/4];
b = [1; 1];
c = [0, 1];
D = 0;
sys4 = ss(A,b,c,D,-1);

% w = linspace(0, 100, 500);
% w = logspace(1,5,500);
w = logspace(-1,2,500);

subplot(2, 1, 1);
h = freqresp(sys1,w);
hmag = abs(squeeze(h));
semilogx(w,hmag)
title('System Magnitude Response for Problem 3.88(a)');
xlabel('Frequency (rads/sec)'); ylabel('Magnitude');
grid on;
subplot(2, 1, 2);
hphase = angle(squeeze(h)) * 180/pi; % convert to degrees
semilogx(w,hphase)
title('System Phase Response for Problem 3.88(a)')
xlabel('Frequency (rads/sec)'); ylabel('Phase (deg)')

```

```

figure
subplot(2, 1, 1);
h = freqresp(sys2,w);
hmag = abs(squeeze(h));
semilogx(w,hmag)
title('System Magnitude Response for Problem 3.88(b)');
xlabel('Frequency (rads/sec)'); ylabel('Magnitude');
grid on;
subplot(2, 1, 2);
hphase = angle(squeeze(h)) * 180/pi; % convert to degrees
semilogx(w,hphase)
title('System Phase Response for Problem 3.88(b)')
xlabel('Frequency (rads/sec)'); ylabel('Phase (deg)')
grid on;

```

```

figure
subplot(2, 1, 1);
h = freqresp(sys3,w);
hmag = abs(squeeze(h));
semilogx(w,hmag)
title('System Magnitude Response for Problem 3.89(a)');
xlabel('Frequency (rads/sec)'); ylabel('Magnitude');
grid on;
subplot(2, 1, 2);
hphase = angle(squeeze(h)) * 180/pi; % convert to degrees
semilogx(w,hphase)
title('System Phase Response for Problem 3.89(a)')
xlabel('Frequency (rads/sec)'); ylabel('Phase (deg)')
grid on;

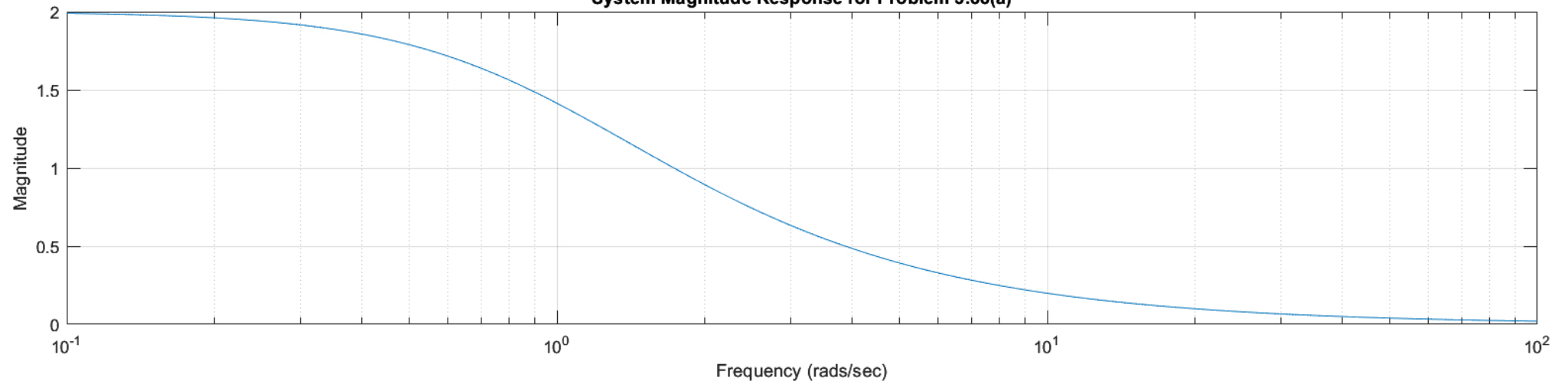
```

```

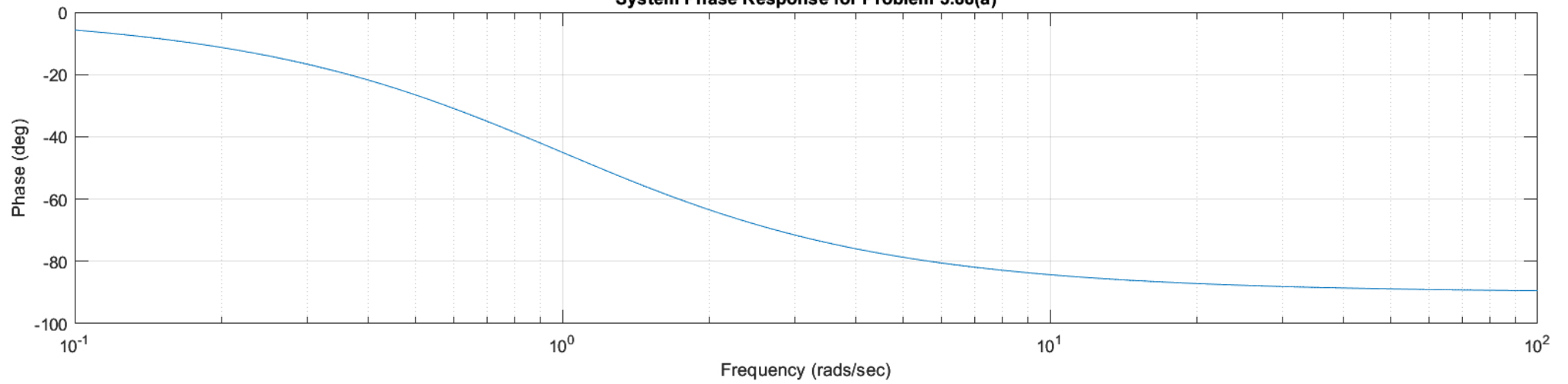
figure
subplot(2, 1, 1);
h = freqresp(sys4,w);
hmag = abs(squeeze(h));
semilogx(w,hmag)
title('System Magnitude Response for Problem 3.89(b)');
xlabel('Frequency (rads/sec)'); ylabel('Magnitude');
grid on;
subplot(2, 1, 2);
hphase = angle(squeeze(h)) * 180/pi; % convert to degrees
semilogx(w,hphase)

```

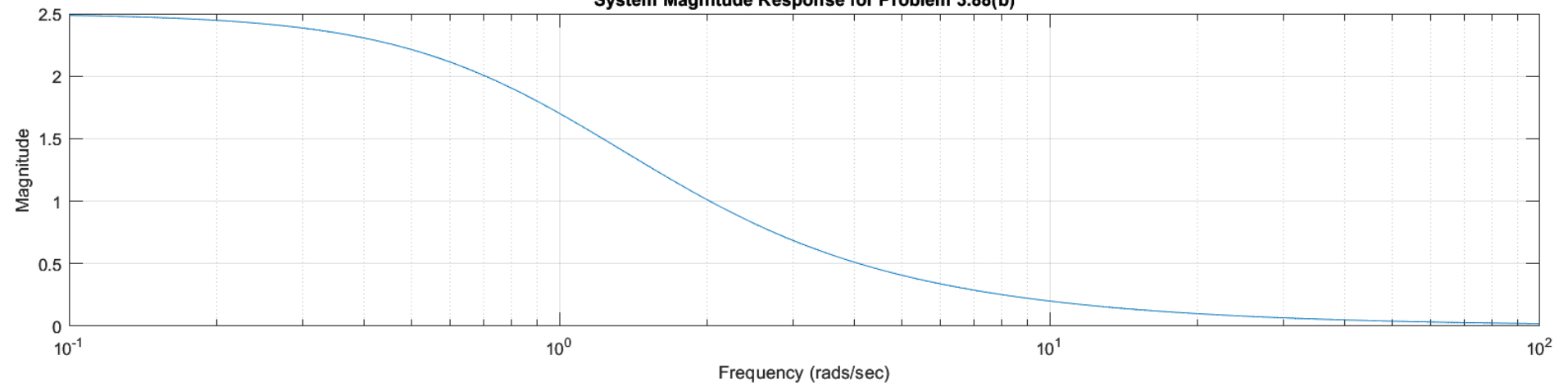
System Magnitude Response for Problem 3.88(a)



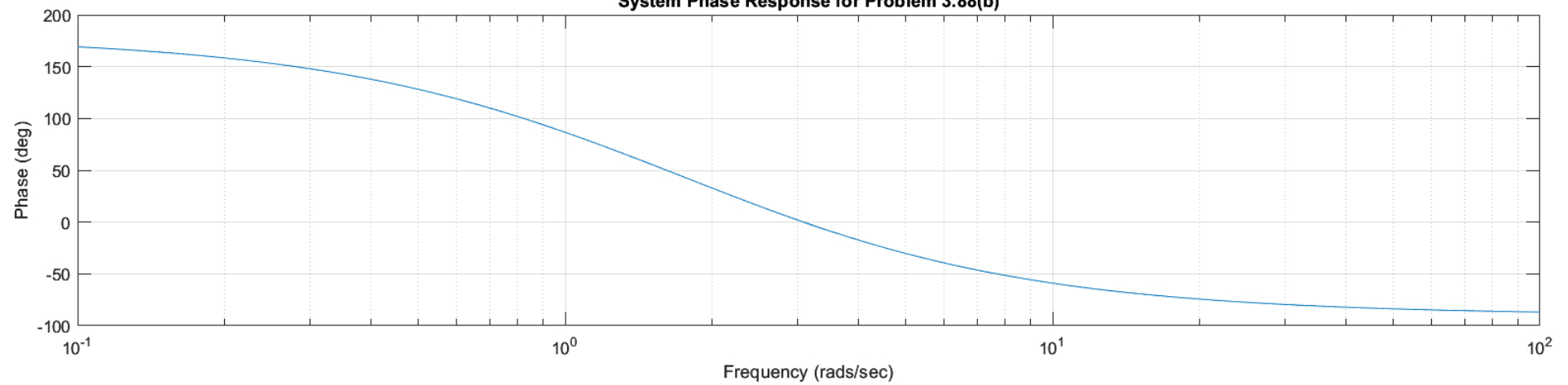
System Phase Response for Problem 3.88(a)



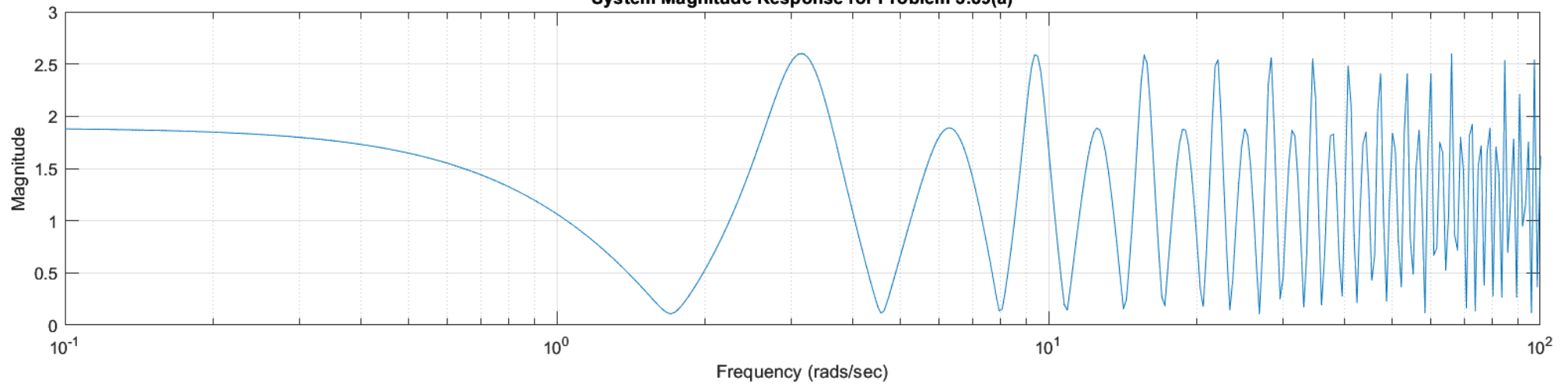
System Magnitude Response for Problem 3.88(b)



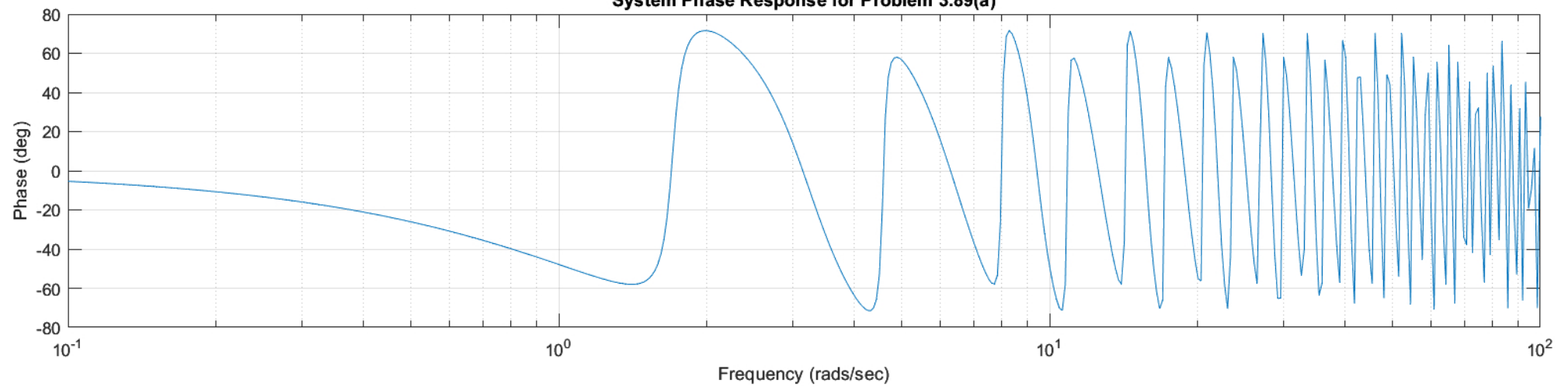
System Phase Response for Problem 3.88(b)



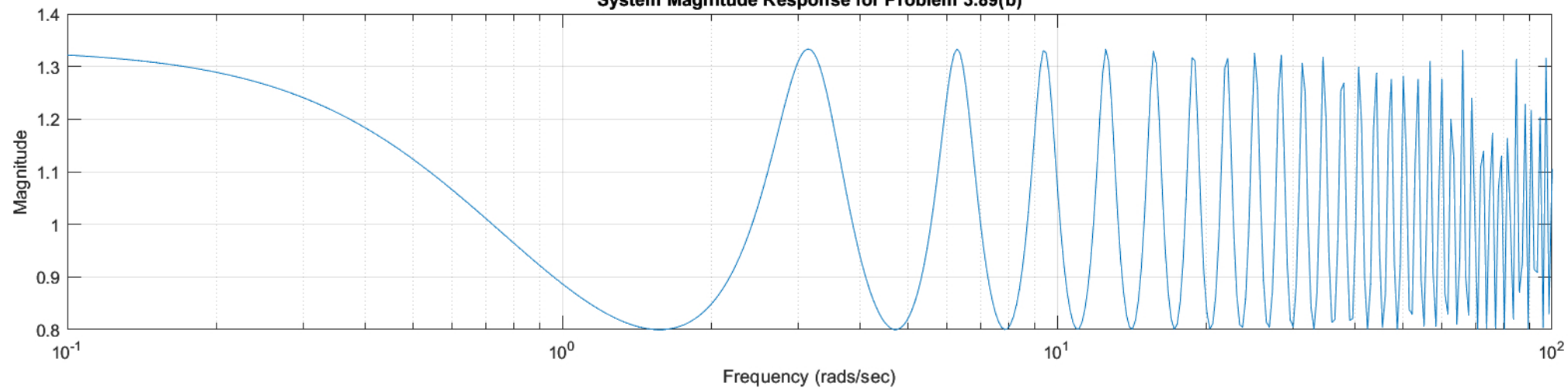
System Magnitude Response for Problem 3.89(a)



System Phase Response for Problem 3.89(a)



System Magnitude Response for Problem 3.89(b)



System Phase Response for Problem 3.89(b)

